Contact

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www.linkedin.com/in/robisen (LinkedIn)
robipsen.wordpress.com/ (Personal)

Top Skills

Artificial Intelligence (AI)
Military History
Encryption

Certifications

R Programming
Improving Deep Neural Networks: Hyperparameter tuning, Regularization and Optimization
Practical Machine Learning

Honors-Awards

Best Paper 2010-2011 School Year

Publications

Android in Action, Second Edition
Unlocking Android: A Developer's Guide
BUILDING AN ANDROID TO-DO APP WITH PHONEGAP, HTML AND JQUERY
Cyber and Information Threats to Seaports and Ships in Maritime Security, Second Edition: An Introduction 2nd Edition by Michael McNicholas
Building smarter wearables for healthcare, Part 2, Designing cognitive applications that take advantage of the Watson services

Patents

Distributed Wireless Communications for Tactical Network Dominance

Robi Sen

CEO
Richland, Washington, United States

Summary

I am an inventor, engineer, and futurist focused on solving the complex challenges of our time through deep, disruptive innovation. My work spans counter-autonomy/AI, biodefense, and advanced electronic warfare systems—reflecting a commitment to advancing technology in defense of national and global security.

In 2008, I co-created Mesmer, a pioneering system capable of safely landing hostile drones, anticipating the Counter-Unmanned Aerial Vehicle (C-UAV) market before it formally existed. This early success reflects my ability to foresee and shape emerging technological frontiers.

My career includes collaborations with DARPA, NASA, DOE, and the Department of Defense, where I've contributed to cutting-edge programs that redefine what's technically and operationally possible. I hold a portfolio of patents and have authored publications that influence the trajectory of defense and dual-use technologies.

I work to give my clients a decisive edge through transformative products, services, and capabilities. If you're ready to build what comes next, let's collaborate.

Experience

Cognoscenti LLC

CEO
December 2018 - Present (7 years 1 month)
Washington State

- Created Cognoscenti to fill the need for tools, services, and specialized hardware for designing, developing, and testing AI-powered Radio Frequency (RF) devices.

System and Method for Performing Frictionless Collaboration for Criteria Search

LPI/LPD Communication Systems

LPI/LPD communication systems

Intrusion detection and radio fingerprint tracking

- Researching novel Counter Unmanned Aircraft Systems (CUAS) exploits and mitigations. Performing both literature analysis and building out unusual attacks of my own design.
- Worked with a small defense company to invent and create intelligent 5G micro cells driven by 'AI.' System can perform tasks such as smart network slicing and channel bonding.
- Currently working on AI based cyber exploitation of radio frequency systems using a series of novel approaches and algorithms to attack the physical layer.
- Invented a new approach to creating Low Probability to Detect and Low Probably to Intercept (LPI/LPD) system.
- Developed a machine learning based optical detection system for government customers drones.
- Assisted Northeastern's Kostas Research Institute in creating a Machine Learning system for RF detection with a focus on Counter Unattended Aerial Vehicles (C-UAS).
- Designed and developed a novel method for Predictive RF Spectroscopy for use in predicting real-time spectrum usage for communications and Electronic Warfare (EW); patent pending.
- Designed a system to convert Blackbox neural networks to human readable set of rules to assist in design and testing of neural networks with a special focus on autonomous vehicles.
- Assisting and supporting a private company in developing their directed energy system for countering UAS.

Department 13, LLC

16 years 11 months

CTO and Founder

September 2015 - December 2018 (3 years 4 months)

Falls Church, VA

Founder, Partner, CSO

February 2002 - November 2015 (13 years 10 months)

Entity-X

CSO and Founder

May 2009 - December 2018 (9 years 8 months)

Twin Technologies

Managing Partner and EIR

July 2007 - August 2014 (7 years 2 months)

Help define and grow the company taking on many rolls.

Granularity Information Architecture
Partner, Vice President of Software Development
August 1998 - February 2002 (3 years 7 months)

Cofounded Granularity and was Vice President of Software.

Grew company from two employees to fifty active workers.

Instituted project management practices and software development practices including the use of UML and Patterns for Web Applications long before these where popular practices.

Worked on internal Intellectual Property and products including the XML tool GIAX which was the first common usage mapping names to XML node for simpler and easier parsing .

Managed multiple large accounts.

Worked directly with customers on complex EAI projects helping integrate Web Applications with ERP’s and low level back office systems.

Championed the company as well as emerging technologies of the time including XML, Web Services, and the like.

Education

Norwich University
Masters, Military History · (2008 - 2010)